XRPL Validator — Daily Engineering Report · 2026-07-23

Period: 2026-07-23 00:00 to 24:00 US Eastern (2026-07-23 04:00 to 2026-07-24 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04); data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-07-23): 1.00000 (0 missed of 21,621). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(xrpl-ledger): enforce owner reserve on object creation — TicketCreate + TrustSet** (`bcf7e5c`, xrpl-validator) The native engine minted owned objects without checking the account could afford the incremental owner reserve, returning tesSUCCESS where mainnet returned tecINSUFFICIENT_RESERVE (TicketCreate) / tecINSUF_RESERVE_LINE (TrustSet). Closes ~22 fresh-traffic scout divergences in one class fix.
- **fix(xrpl-ledger): AMMDeposit owner-reserve guard for zero-LPToken depositors** (`385e129`, xrpl-validator) Same owner-reserve class as the TicketCreate/TrustSet fix: an AMMDeposit by an account holding ZERO LPTokens opens (or funds) an LPToken trust line and so must keep XRP above accountReserve(ownerCount + 1). The native engine skipped this and over-applied (tesSUCCESS, 4-5 muts) where mainnet returned tecINSUF_RESERVE_LINE (net_muts=1).
- **fix(xrpl-ledger): direct IOU payment requires an existing destination trust line** (`a6ef95a`, xrpl-validator) A Payment delivers an IOU by rippling through the issuer, which credits the destination's EXISTING trust line — rippled never opens a trust line for the receiver. Our apply_iou_direct phantom-created the destination's line and delivered, returning tesSUCCESS (with extra RippleState + owner-dir + dir-page mutations) where mainnet returned tecPATH_DRY (fee-only), because the destination did not trust the issuer.
- **fix(xrpl-ledger): payments never autobridge — AMM over-mutation via a synthesized XRP leg** (`4a36c48`, xrpl-validator) rippled synthesizes the XRP bridge only in FlowCross (offer crossing). A payment's strands come from its own Paths plus the default path, so it reaches a book only where its path names one. cross_engine_to bridged unconditionally for IOU<->IOU, inventing a route the transaction never asked for.

- **fix(xrpl-ledger): no AMM liquidity in permissioned-domain books — corpus 100%** (`84ced29`, xrpl-validator) BookStep::tryAMM returns early when book_.domain is set ("amm doesn't support domain yet", BookStep.cpp:820), so a pool never participates in a domain-scoped book. cross_engine_to discovered the AMM regardless of domain, letting a domain offer cross a pool that rippled does not admit.
- **fix(xrpl-ledger): a payment cannot open the destination's trust line** (`e27d088`, xrpl-validator) Every payment strand ends by rippling the delivered IOU from its issuer to the destination, and that final step is a DirectStepI whose check opens with "Since this is a payment a trust line must be present" — no line yields terNO_LINE (DirectStep.cpp:423). Because every strand shares that same last step, one missing line invalidates all of them, and a flow with no surviving strands is tecPATH_DRY.
- **fix(xrpl-ledger): charge the issuer's TransferRate on payment delivery** (`2b1beae`, xrpl-validator) The engine ignored TransferRate entirely, so every payment that moved an IOU between two non-issuers delivered the gross amount. rippled charges the issuer's rate on the strand's last step: DirectStepI issues to the destination and qualitiesSrcIssues sets srcQOut = transferRate(issuer) whenever the previous step redeems (DirectStep.cpp:765), which for a payment is always — BookStep reports DebtDirection::Redeems …
- **fix(xrpl-ledger): AccountDelete fee is one owner reserve, read from the ledger** (`3cebd6a`, xrpl-validator) AccountDelete::calculateBaseFee returns calculateOwnerReserveFee(view, tx) — "the fee required for AccountDelete is one owner reserve". That comes from the ledger's fee settings and is not a constant: the 2024 vote cut the increment from 2 XRP to 0.2, so a hardcoded 2_000_000 rejects every present-day AccountDelete.
- **fix(xrpl-ledger): merging past an oracle's ten-entry cap is tecARRAY_TOO_LARGE** (`bfc83f9`, xrpl-validator) OracleSet is an upsert. Overflowing kMaxOracleDataSeries by MERGING the transaction's series into the existing object is a tec, not a tem: the transaction is well formed and only the resulting object is too big, so the fee is claimed (OracleSet.cpp:168). temARRAY_TOO_LARGE is a separate preflight check on the transaction's own array (line 46), which is not what these hit.
- **fix(xrpl-ledger): correct the credential keylet, allow self-issuance, hydrate it** (`08515b3`, xrpl-validator) Four faults stacked on one transaction, each hiding the next.
- **fix(xrpl-ledger): tecOBJECT_NOT_FOUND for missing NFT offers, tecAMM_BALANCE for over-withdraw** (`ce27e48`, xrpl-validator) Two unrelated result-code gaps, both fee-only on mainnet.
- **feat(xrpl-ledger): DX_BOOK trace for the book walk's quality gate** (`eb19a38`, xrpl-validator) Prints each directory's encoded quality against the taker's threshold at the point the walk decides to break, in the same style as DX_AMM and DX_BRIDGE. The crossing gate is a 64-bit encoded-rate comparison where a single ulp decides whether an offer is reachable, and there was no way to see either side of it without a rebuild.
- **fix(xrpl-ledger): reject offers by achieved quality and skip expired ones; keep the cleanup on tecKILLED** (`a823d98`, xrpl-validator) Three faults in book crossing, uncovered together on two IoC offers that mainnet leaves completely untouched.

Tests and campaigns

- **test(xrpl-node): repeatable corpus sweep + restore the #105761560 fixture** (`6b3b87b`, xrpl-validator)

Independence program

No independence record for UTC day 2026-07-23; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Expiring Trust Lines with Reserve Sweep** (tokens) Add an optional sfExpiration (LedgerIndex) SField to RippleState, settable by either side via a new tfSetExpiration flag on TrustSet only when that si
- **Auditor-Keyed Disclosure for Confidential MPT** (privacy) Extend the confidential MPT transfer transaction (per XLS-96) with an optional sfAuditCiphertext blob (bounded to 512 bytes), populated only when the
- **Canonical Wrapped-Asset Object for XChainBridge** (interop) Add a new ltWRAPPEDASSET ledger object (keylet::wrappedAsset(bridgeID, issuer)) auto-created on a Bridge's first XChainCommit, tracking sfOutstandingS

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
07-23 15:28	—	xrpl-validator	`bcf7e5c`	fix(xrpl-ledger): enforce owner reserve on object creation — TicketCreate + TrustSet
07-23 16:03	—	xrpl-validator	`385e129`	fix(xrpl-ledger): AMMDeposit owner-reserve guard for zero-LPToken depositors
07-23 20:37	—	xrpl-validator	`a6ef95a`	fix(xrpl-ledger): direct IOU payment requires an existing destination trust line
07-23 23:06	—	xrpl-validator	`4a36c48`	fix(xrpl-ledger): payments never autobridge — AMM over-mutation via a synthesized XRP leg

Committed	Merged into main	Repo	Commit	Subject
07-23 23:06	—	xrpl-validator	`6b3b87b`	test(xrpl-node): repeatable corpus sweep + restore the #105761560 fixture
07-23 23:37	—	xrpl-validator	`84ced29`	fix(xrpl-ledger): no AMM liquidity in permissioned-domain books — corpus 100%
07-24 00:20	—	xrpl-validator	`e27d088`	fix(xrpl-ledger): a payment cannot open the destination's trust line
07-24 01:29	—	xrpl-validator	`2b1beae`	fix(xrpl-ledger): charge the issuer's TransferRate on payment delivery
07-24 02:34	—	xrpl-validator	`3cebd6a`	fix(xrpl-ledger): AccountDelete fee is one owner reserve, read from the ledger
07-24 02:35	—	xrpl-validator	`bfc83f9`	fix(xrpl-ledger): merging past an oracle's ten-entry cap is tecARRAY_TOO_LARGE
07-24 02:35	—	xrpl-validator	`08515b3`	fix(xrpl-ledger): correct the credential keylet, allow self-issuance, hydrate it
07-24 03:05	—	xrpl-validator	`ce27e48`	fix(xrpl-ledger): tecOBJECT_NOT_FOUND for missing NFT offers, tecAMM_BALANCE for over-withdraw
07-24 03:10	—	xrpl-validator	`eb19a38`	feat(xrpl-ledger): DX_BOOK trace for the book walk's quality gate
07-24 03:54	—	xrpl-validator	`a823d98`	fix(xrpl-ledger): reject offers by achieved quality and skip expired ones; keep the cleanup on tecKILLED

Times in UTC. A commit appears on the day it was committed to main or merged into it.